

Task 2 – Network Security & Scanning

Internship Program

ApexPlanet Software Pvt. Ltd.

Submitted By

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Objective

The objective of this task was to learn and perform network reconnaissance, port scanning, vulnerability assessment, network traffic analysis, and basic firewall configuration using industry-standard cybersecurity tools such as Nmap, Nessus Essentials, Wireshark, hping3, and iptables.

1. Environment Setup

Tools Used

- ⑩ Kali Linux
- ⑩ Metasploitable2
- ⑩ Oracle VirtualBox
- ⑩ Nmap
- ⑩ Nessus Essentials
- ⑩ Wireshark
- ⑩ hping3
- ⑩ iptables

Virtual Machines

Kali Linux

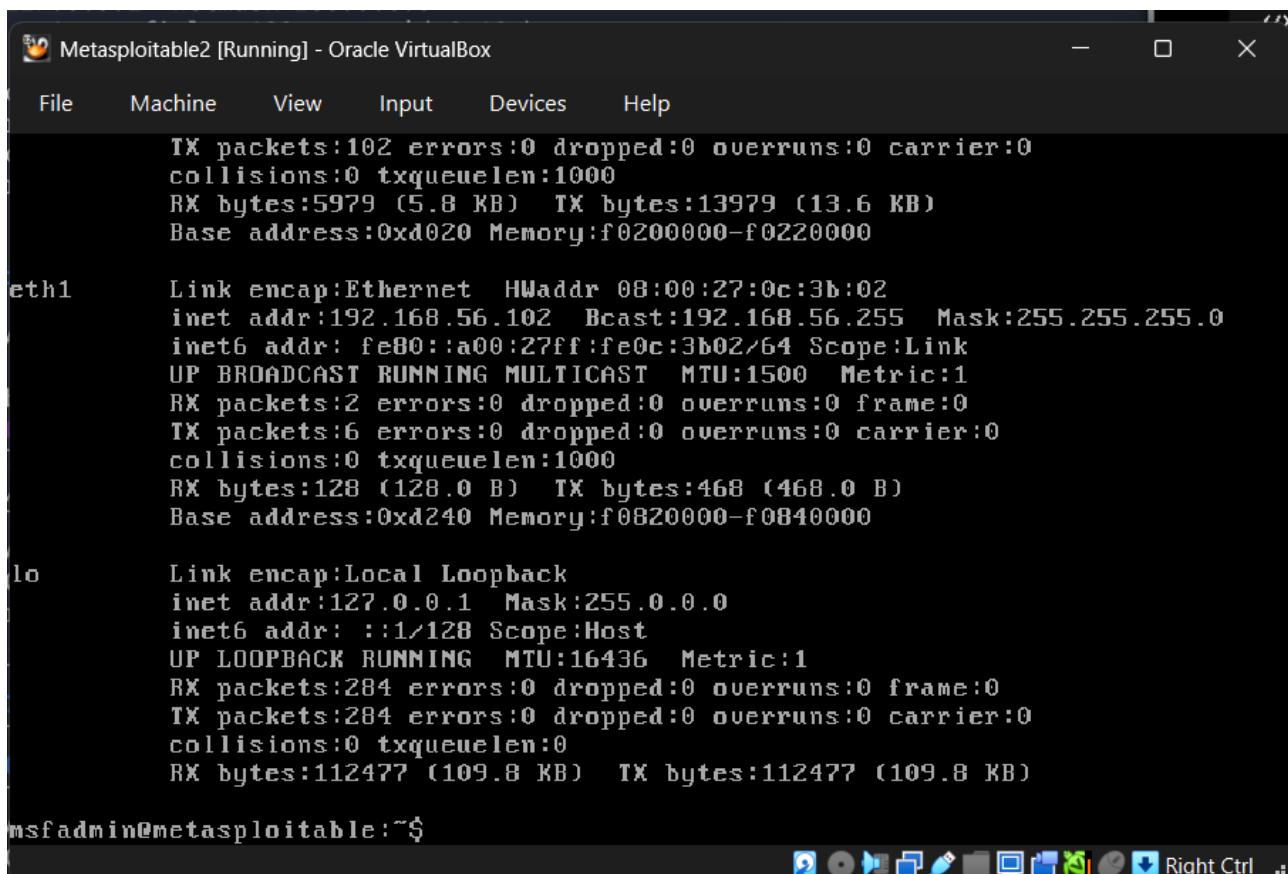
Used as the attacker/security testing machine.

Metasploitable2

Used as the intentionally vulnerable target machine.

```
(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
  ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
  roup default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86311sec preferred_lft 86311sec
    inet6 fd17:625c:f037:2:1abb:8a50:53d9:3b91/64 scope global dynamic nopref
  ixroute
        valid_lft 86185sec preferred_lft 14185sec
    inet6 fe80::2218:9cb7:2586:8421/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
  roup default qlen 1000
    link/ether 08:00:27:62:f1:14 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixro
  ute eth1
        valid_lft 511sec preferred_lft 511sec
    inet6 fe80::57b7:c607:449:2df0/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

VirtualBox network configuration showing Kali Linux and Metasploitable2 connected through Host-Only Adapter.



```
Metasploitable2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

TX packets:102 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:5979 (5.8 KB) TX bytes:13979 (13.6 KB)
Base address:0xd020 Memory:f0200000-f0220000

eth1 Link encap:Ethernet HWaddr 08:00:27:0c:3b:02
      inet addr:192.168.56.102 Bcast:192.168.56.255 Mask:255.255.255.0
      inet6 addr: fe80::a00:27ff:fe0c:3b02/64 Scope:Link
      UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
      RX packets:2 errors:0 dropped:0 overruns:0 frame:0
      TX packets:6 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:1000
      RX bytes:128 (128.0 B) TX bytes:468 (468.0 B)
      Base address:0xd240 Memory:f0820000-f0840000

lo Link encap:Local Loopback
    inet addr:127.0.0.1 Mask:255.0.0.0
    inet6 addr: ::1/128 Scope:Host
    UP LOOPBACK RUNNING MTU:16436 Metric:1
    RX packets:284 errors:0 dropped:0 overruns:0 frame:0
    TX packets:284 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:0
    RX bytes:112477 (109.8 KB) TX bytes:112477 (109.8 KB)

msfadmin@metasploitable:~$
```

Metasploitable2 network interface configuration displaying target IP address 192.168.56.102.

2. Network Reconnaissance

Objective

To identify the target system and verify network connectivity between Kali Linux and Metasploitable2.

Steps Performed

Identify IP Address of Metasploitable2

Command:

```
ifconfig -a
```

Result:

Metasploitable2 IP Address:

192.168.56.102

Verify Connectivity

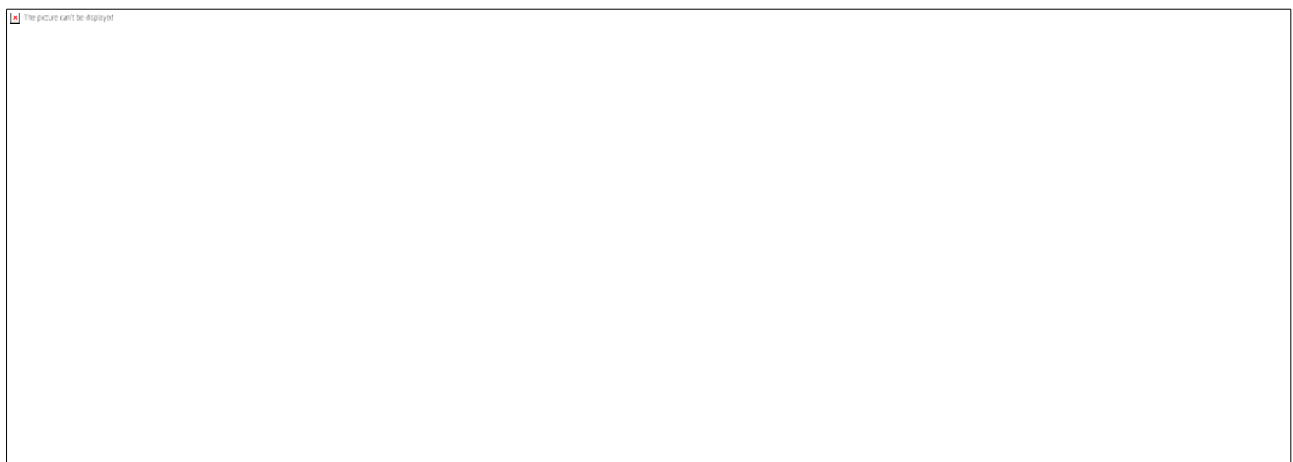
Command:

```
ping 192.168.56.102 -c 4
```

Observation:

The target responded successfully to all ICMP packets.

Successful ICMP connectivity test from Kali Linux to Metasploitable2.



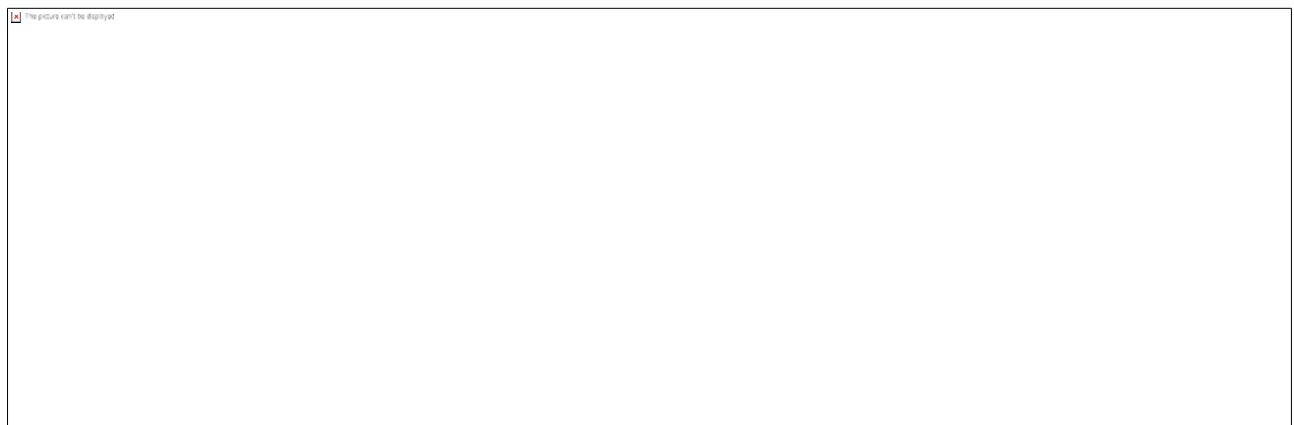
WHOIS passive reconnaissance.

```
(kali㉿kali)-[~]
$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-06-06T16:03:03Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
```

DNS lookup using nslookup.



Banner grabbing to identify running services.



3. Port and Service Scanning

Objective

To discover open ports and services running on the target machine.

TCP Scan

Command:

```
sudo nmap -sS 192.168.56.102
```

Observation:

Multiple ports were found open including:

- 🔩 21 (FTP)
- 🔩 22 (SSH)
- 🔩 23 (Telnet)
- 🔩 25 (SMTP)
- 🔩 53 (DNS)
- 🔩 80 (HTTP)
- 🔩 111 (RPCBind)
- 🔩 139 (NetBIOS)
- 🔩 445 (SMB)
- 🔩 3306 (MySQL)
- 🔩 5432 (PostgreSQL)

 The picture can't be displayed

UDP Scan

Command:

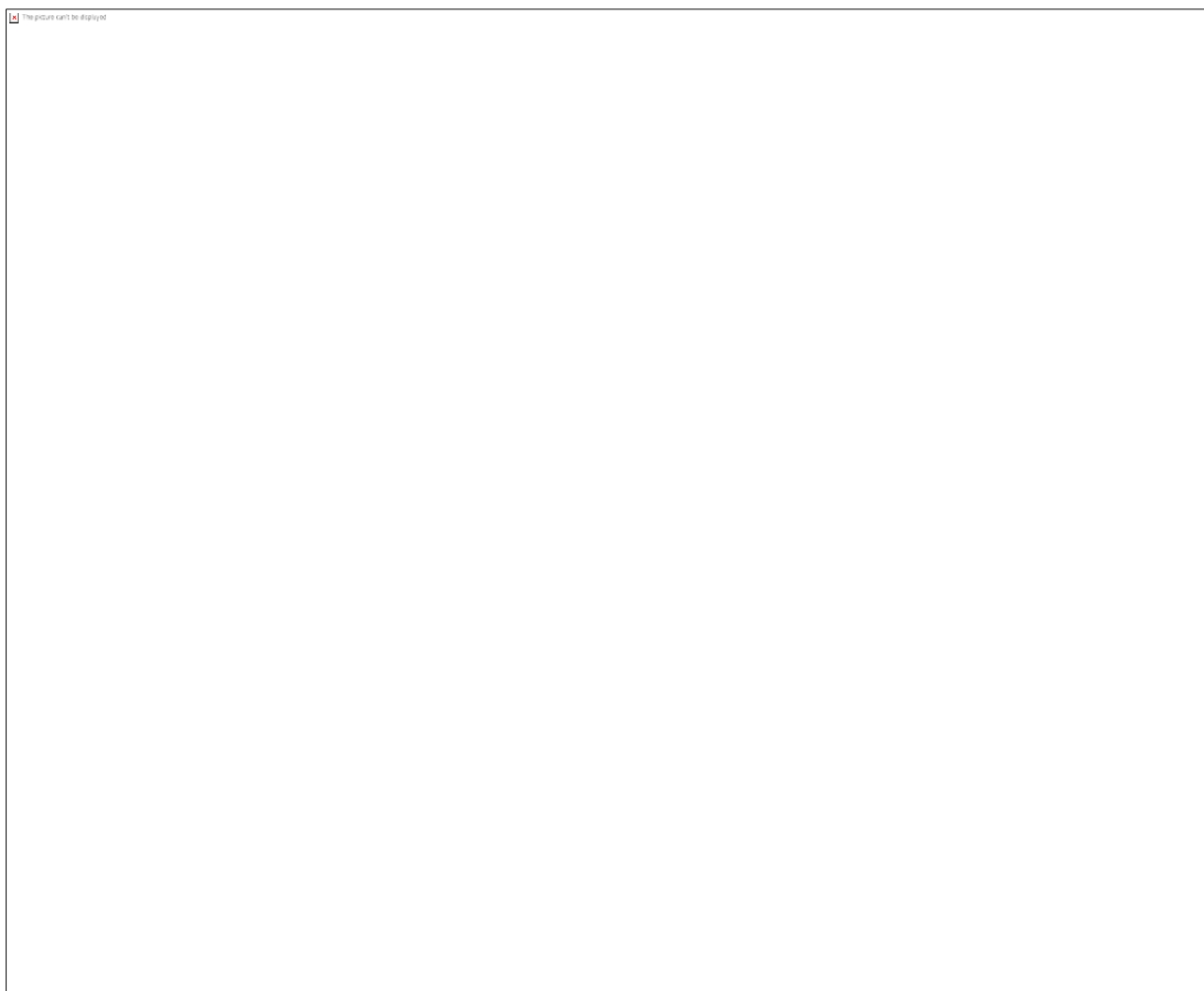
```
sudo nmap -sU 192.168.56.102
```

Observation:

Open UDP services identified:

- 🔩 DNS (53)
- 🔩 RPCBind (111)
- 🔩 NetBIOS (137)
- 🔩 NFS (2049)

```
(kali㉿kali)-[~]  
$ sudo nmap -sU 192.168.56.102  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-06 12:17 -0400  
Stats: 0:10:51 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 64.89% done; ETC: 12:34 (0:05:51 remaining)  
Stats: 0:10:53 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 65.09% done; ETC: 12:34 (0:05:49 remaining)  
Stats: 0:10:54 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 65.19% done; ETC: 12:34 (0:05:48 remaining)  
Stats: 0:10:55 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 65.29% done; ETC: 12:34 (0:05:47 remaining)  
Stats: 0:10:56 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 65.39% done; ETC: 12:34 (0:05:46 remaining)  
Stats: 0:10:58 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 65.59% done; ETC: 12:34 (0:05:44 remaining)
```



Service Version Detection

Command:

```
sudo nmap -sV 192.168.56.102
```

Observation:

Detected versions of running services such as:

- ⑩ vsFTPD 2.3.4
- ⑩ OpenSSH 4.7p1
- ⑩ Apache HTTP Server 2.2.8
- ⑩ MySQL
- ⑩ PostgreSQL

```
(kali㉿kali)-[~]
$ nmap -sV 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-06 12:09 -0400
Nmap scan report for 192.168.56.102
Host is up (0.093s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rrexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:0C:3B:02 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix,
Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.06 seconds
```

Operating System Detection

Command:

```
sudo nmap -O 192.168.56.102
```

Observation:

The target operating system was identified as Linux Kernel 2.6.x.


```
Nmap scan report for 192.168.56.102
Host is up (0.029s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:0C:3B:02 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submi
t/ .
Nmap done: 1 IP address (1 host up) scanned in 6.35 seconds
```

Aggressive Scan

 The picture can't be displayed

 The picture can't be displayed

4. Nmap Report Generation

Objective

Generate a scan report for documentation.

Command:

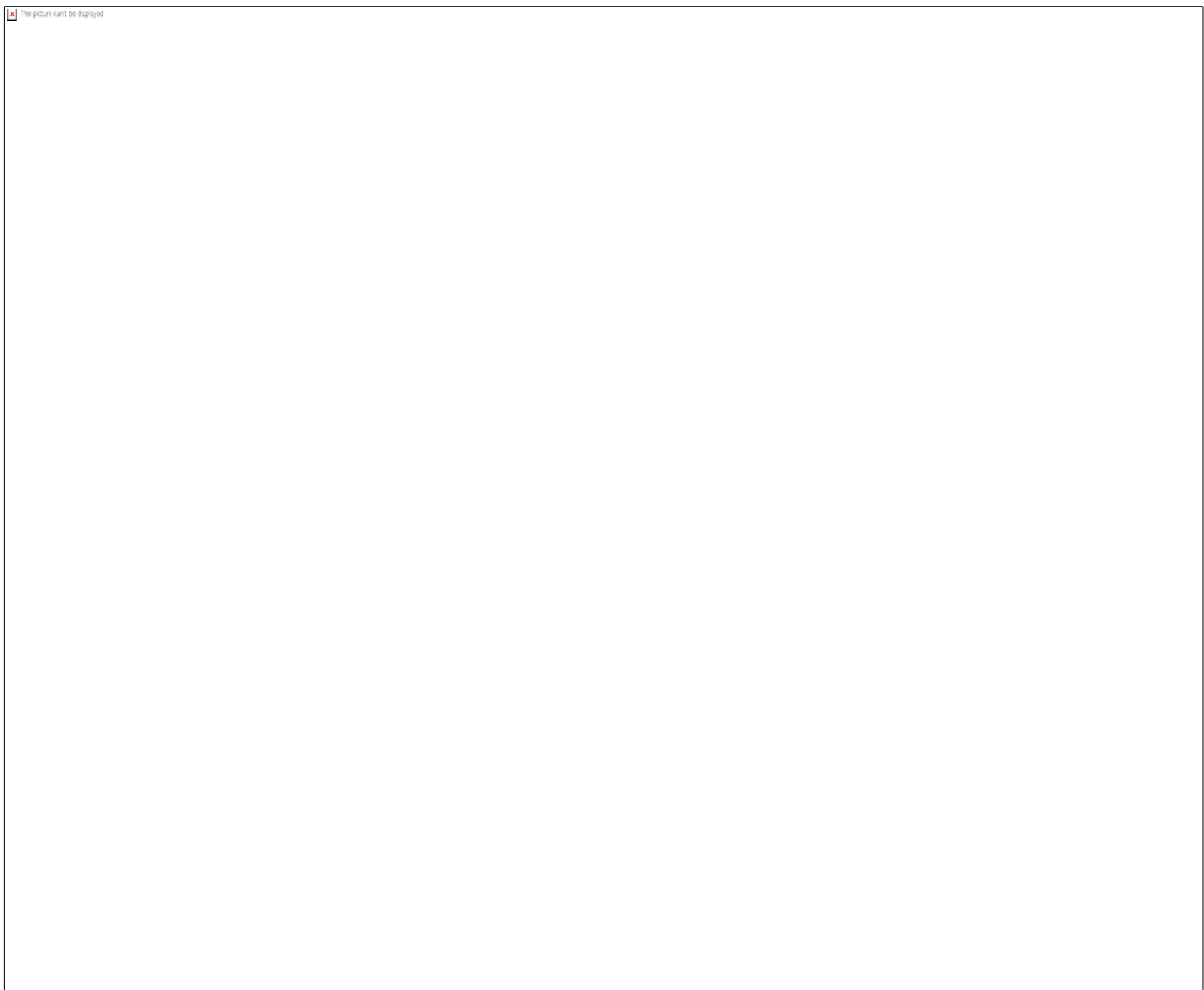
```
nmap -A 192.168.56.102 -oN nmap_report.txt
```

Observation:

A detailed report containing open ports, services, operating system details, and scripts output was successfully generated.

 The picture can't be displayed

```
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ ssl-date: 2026-06-06T06:36:22+00:00; -10h01m42s from scanner time.
|_ smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, S
TARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
| sslv2:
|   SSLv2 supported
|   ciphers:
|   SSL2_DES_64_CBC_WITH_MD5
|   SSL2_DES_192_EDE3_CBC_WITH_MD5
|   SSL2_RC4_128_WITH_MD5
|   SSL2_RC2_128_CBC_WITH_MD5
|   SSL2_RC4_128_EXPORT40_WITH_MD5
|_  SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/sta
teOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_ Not valid after:  2010-04-16T14:07:45
53/tcp    open  domain       ISC BIND 9.4.2
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_ http-title: Metasploitable2 - Linux
111/tcp    open  rpcbind      2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000   2                111/tcp    rpcbind
|   100000   2                111/udp    rpcbind
|   100003   2,3,4           2049/tcp   nfs
```



```
(kali㉿kali)-[~]  
$ ls  
192.168.56.102  Documents  internship_test  nmap_report.txt  Templates  
decrypted.txt   Downloads  message.txt     Pictures          Videos  
Desktop        encrypted.enc Music           Public
```

5. Vulnerability Assessment using Nessus Essentials

Objective

Identify security vulnerabilities present on the target system.

Nessus Installation

Steps:

1. Download Nessus Essentials.
2. Install using dpkg.

3. Start Nessus service.

4. Access Nessus Web Interface.

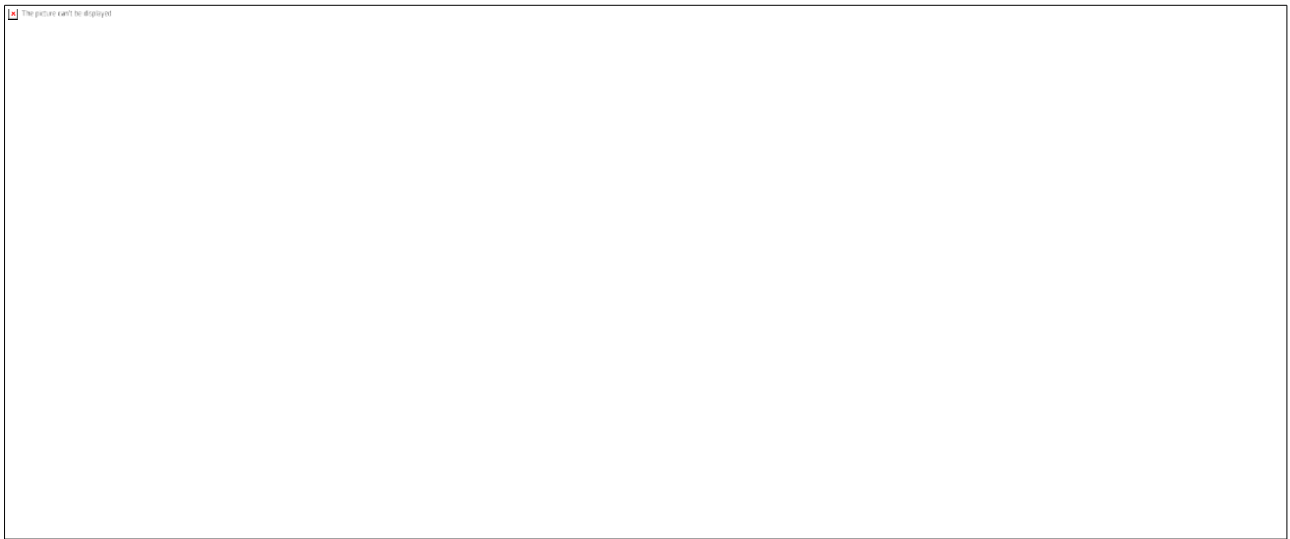
Commands:

```
sudo dpkg -i Nessus*.deb
```

```
sudo systemctl start nessusd
```

```
sudo systemctl status nessusd
```

Nessus Installation



Nessus Setup

 The picture can't be displayed

Nessus Essentials Activation



Welcome to Nessus

Choose how you want to deploy Nessus. Select an option to get started.

- ☐ Set up a purchased instance of Nessus
- ☐ Start a trial of Nessus Expert
- ☐ Start a trial of Nessus Professional
- ☐ Register for Nessus Essentials
- ☒ Set up Nessus Essentials Plus for Education
- ☐ Link Nessus to another Tenable product

Back

Continue

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Plugin Download



Initializing

Please wait while Nessus is initializing.

Downloading plugins...



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Vulnerability Scan

Target:

192.168.56.102

Scan Type:

Basic Network Scan

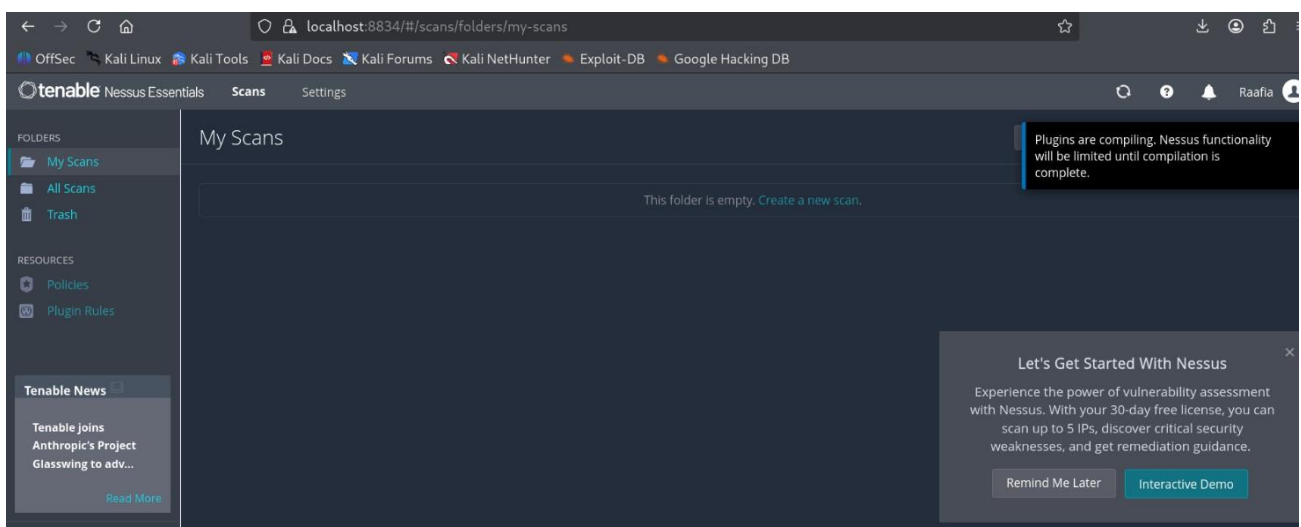
Observation:

Nessus identified multiple vulnerabilities of varying severity levels.

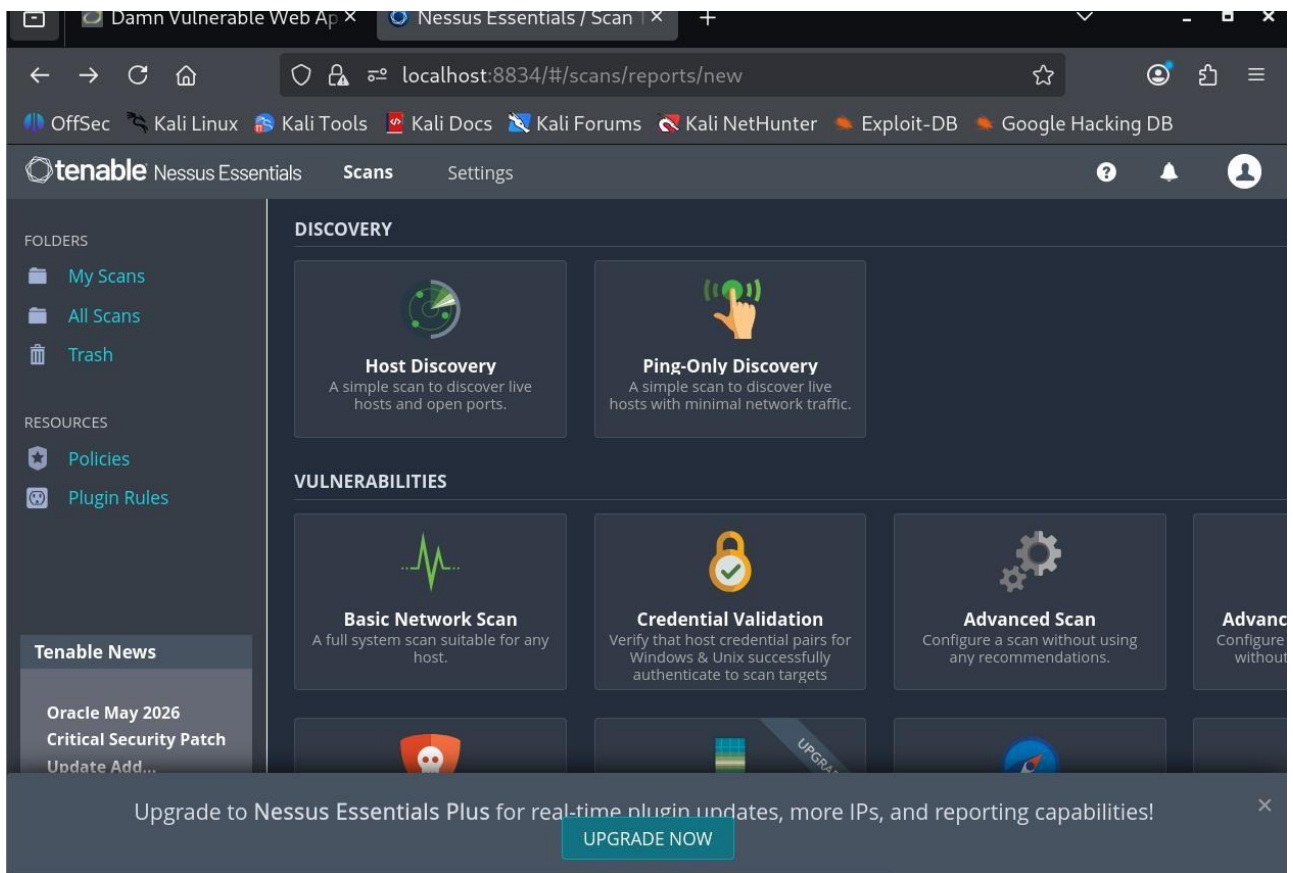
Categories included:

- ⑩ Critical
- ⑩ High
- ⑩ Medium
- ⑩ Low
- ⑩ Informational

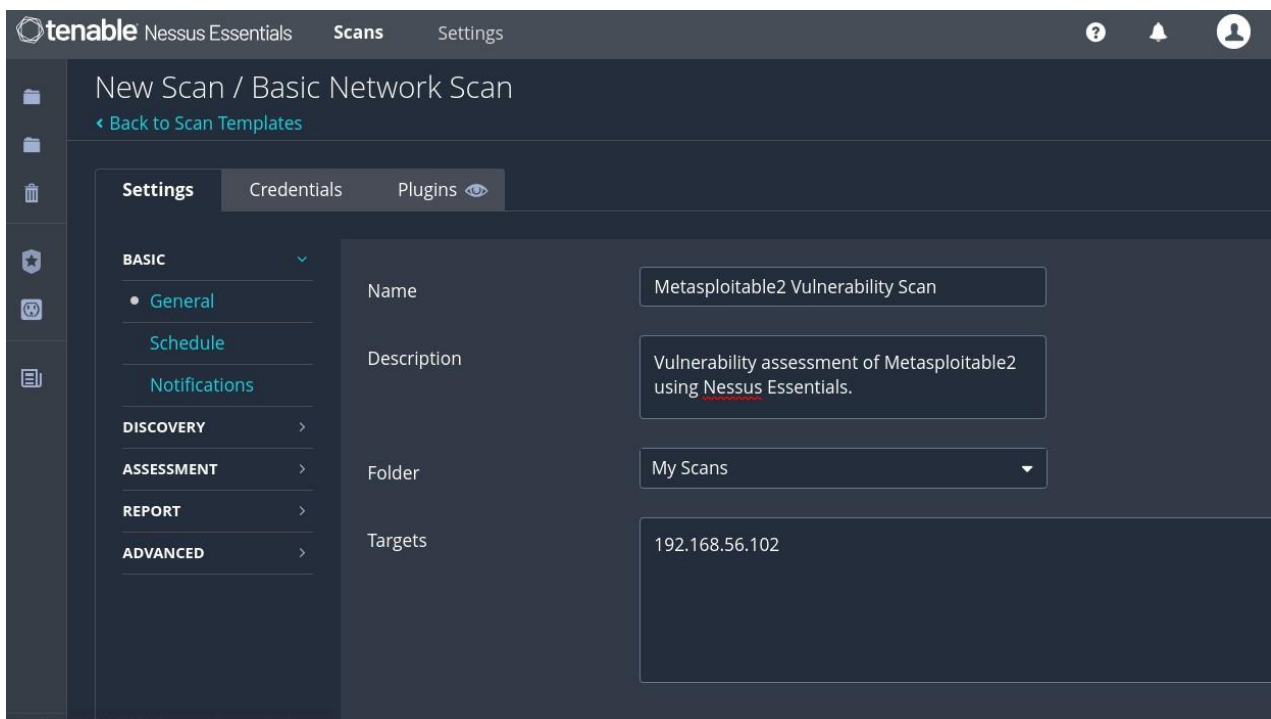
Nessus Dashboard



Scan Template



Scan Configuration



Running Scan

My Scans

Import

New Folder

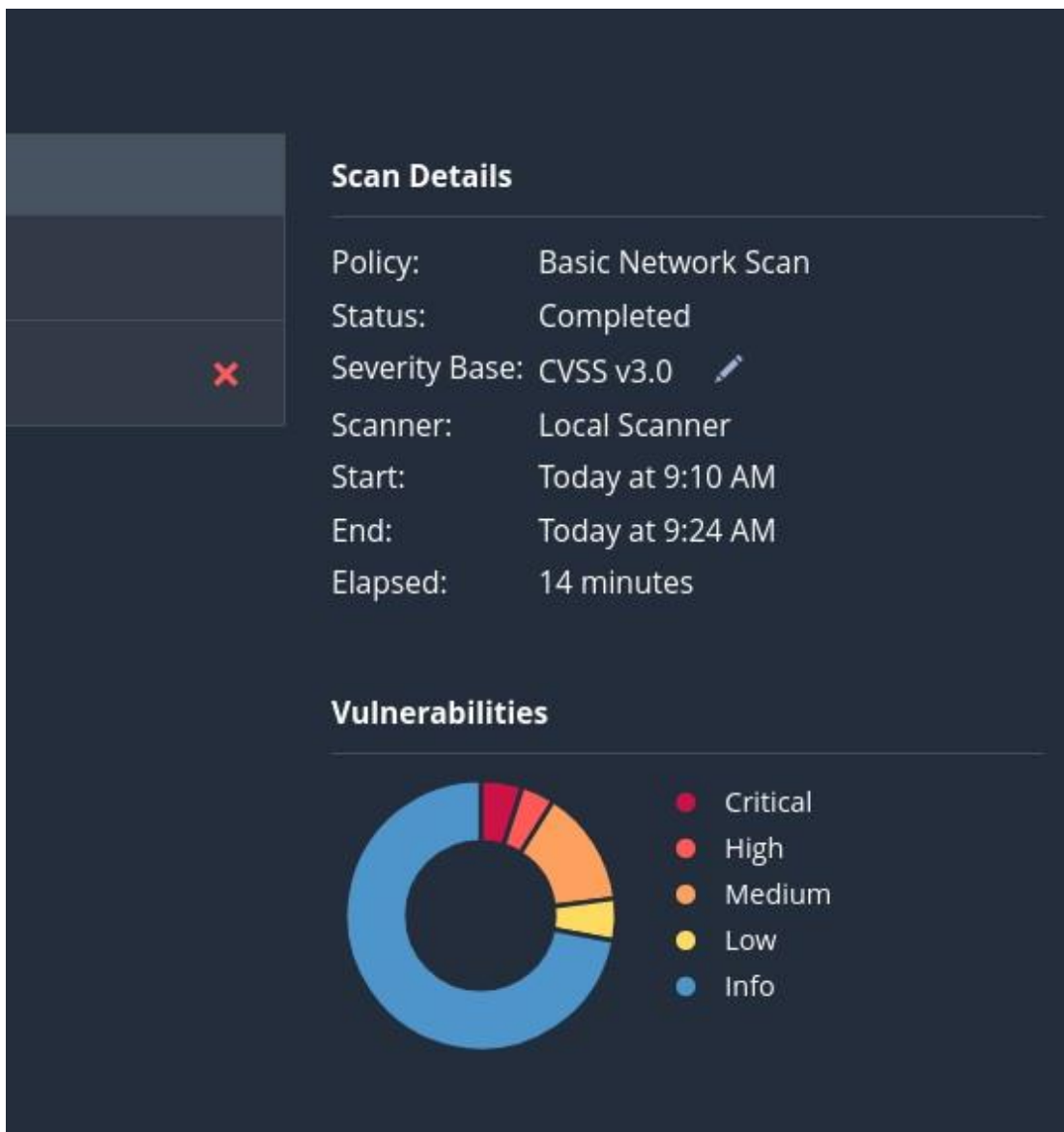
New Scan

Search Scans

1 Scan

☐	Name	Scan Type	Schedule	Last Scanned
☐	Metasploitable2 Vulnerability Scan	Vulnerability	On Demand	Today at 9:10 AM

Results Overview



Vulnerabilities List

Metasploitable2 Vulnerability Scan Configure Audit Trail

Back to My Scans

Hosts 1

Vulnerabilities 71

Remediations 3

History 2

Filter

Search Vulnerabilities

71 Vulnerabilities

☐	Sev	CVSS	VPR	EPSS	Name	Family	Count	
☐	CRITICAL	10.0			Canonical Ubuntu Linux SEoL (8.04.x)	General	1	
☐	CRITICAL	10.0 *			UnrealIRCd Backdoor Detection	Backdoors	1	
☐	CRITICAL	10.0 *			VNC Server 'password' Password	Gain a shell remotely	1	
☐	CRITICAL	9.8			SSL Version 2 and 3 Protocol Detection	Service detection	2	
☐	CRITICAL	9.8			Bind Shell Backdoor Detection	Backdoors	1	
☐	CRITICAL	...	...	...	SSL (Multiple Issues)	Gain a shell remotely	3	
☐	HIGH	7.5			NFS Shares World Readable	RPC	1	
☐	HIGH	7.5 *			rlogin Service Detection	Service detection	1	
☐	HIGH	7.5 *			rsh Service Detection	Service detection	1	
☐	HIGH	7.5			Samba Badlock Vulnerability	General	1	
☐	MIXED	...	...	...	SSL (Multiple Issues)	General	28	
☐	MIXED	...	...	...	ISC Bind (Multiple Issues)	DNS	5	
☐	MEDIUM	6.5			TLS Version 1.0 Protocol Detection	Service detection	2	
☐	MEDIUM	6.5			Unencrypted Telnet Server	Misc.	1	

Vulnerability Details

Metasploitable2 Vulnerability Scan / Plugin #201352 Configure Audit Trail Launch Report Export

Back to Vulnerabilities

Hosts 1

Vulnerabilities 71

Remediations 3

History 2

CRITICAL Canonical Ubuntu Linux SEoL (8.04.x)

< >

Plugin Details

Description

According to its version, Canonical Ubuntu Linux is 8.04.x. It is, therefore, no longer maintained by its vendor or provider.

Lack of support implies that no new security patches for the product will be released by the vendor. As a result, it may contain security vulnerabilities.

Solution

Upgrade to a version of Canonical Ubuntu Linux that is currently supported.

See Also

<http://www.nessus.org/u73bdb2d2e>

Output

```
OS : Ubuntu Linux 8.04
Security End of Life : May 9, 2013
Time since Security End of Life (Est.) : >= 13 years
```

To see debug logs, please visit individual host

Port	Hosts
80 / tcp / www	192.168.56.102

Risk Information

Risk Factor: Critical

CVSS v3.0 Base Score: 10.0

CVSS v3.0 Vector: CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H

CVSS v2.0 Base Score: 10.0

CVSS v2.0 Vector: CVSS2#AV:N/AC:L/Au:N/C:C/I:C/A:C

Vulnerability Information

CPE: cpe:/o:canonical:ubuntu_linux

Unsupported by vendor: true

6. Packet Analysis using Wireshark

Objective

Analyze different types of network traffic and understand protocol behavior.

Interface Selection



Capture Started

 The picture can't be displayed.

HTTP Traffic Analysis

Procedure:

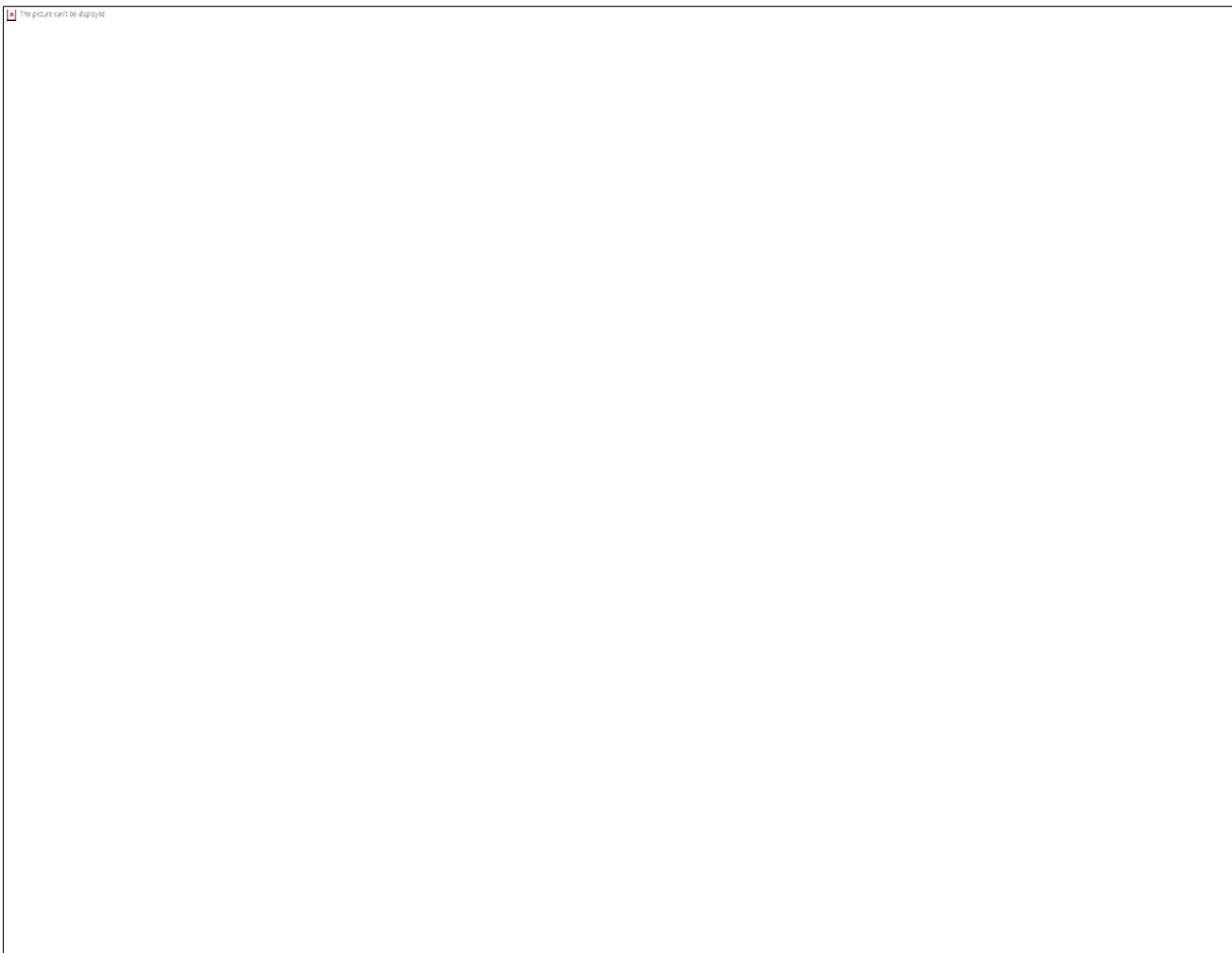
1. Started Wireshark capture.
2. Opened target web application.
3. Captured HTTP packets.

Filter:

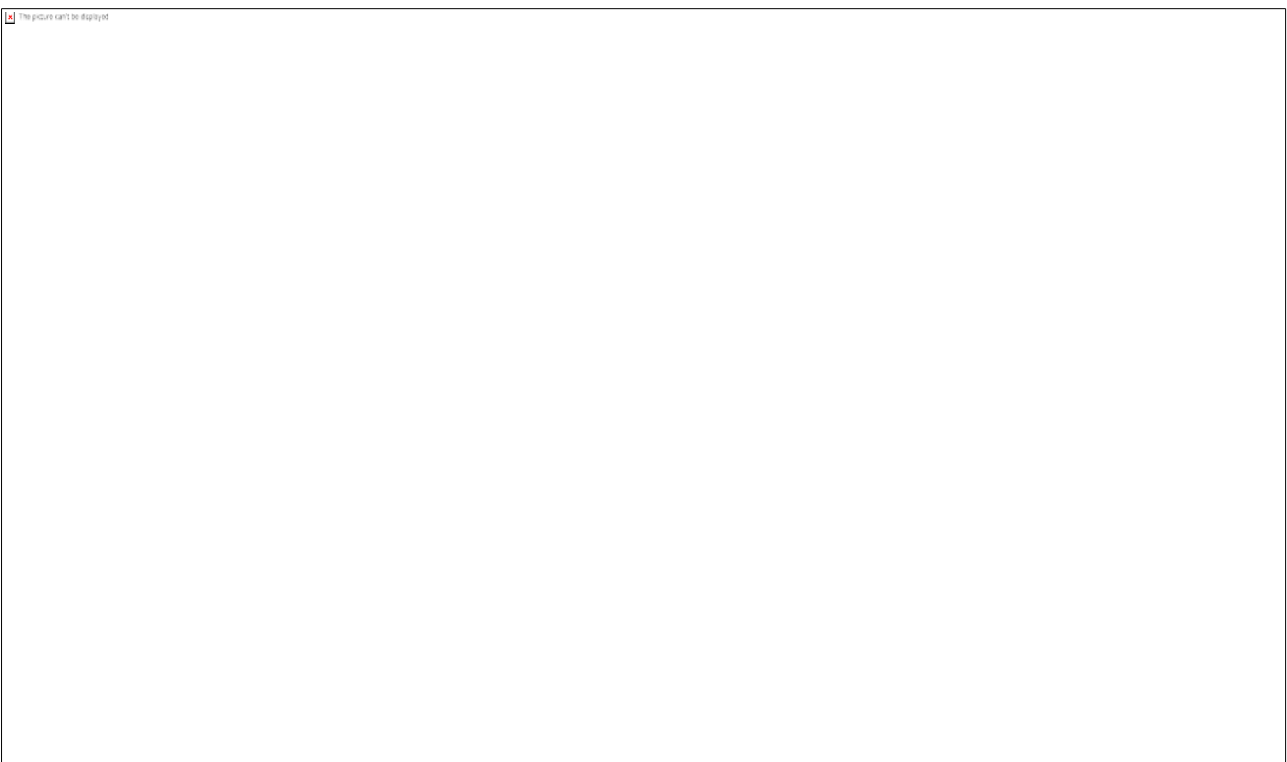
http

Observation:

HTTP requests and responses were successfully captured and analyzed.



HTTP GET and response traffic.



Detailed HTTP packet inspection.

DNS Traffic Analysis

Procedure:

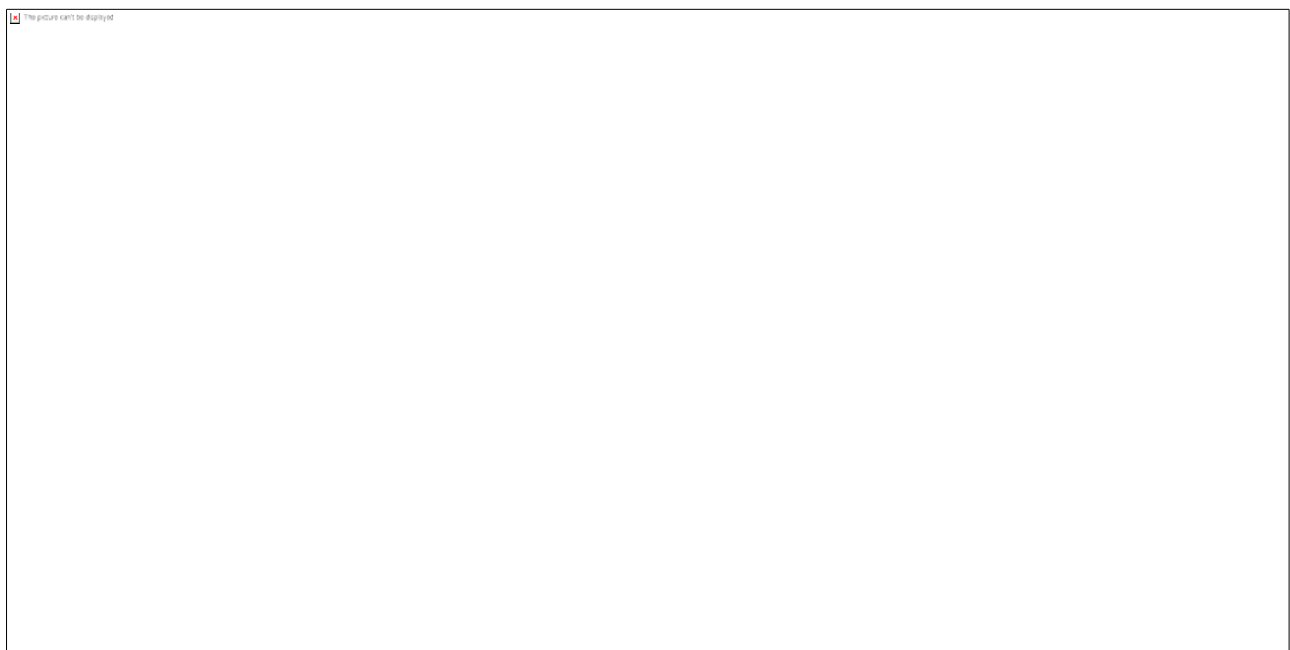
1. Started Wireshark capture.
2. Generated DNS queries.
3. Captured DNS packets.

Filter:

dns

Observation:

DNS query and response packets were observed.



DNS query/response traffic.

The picture can't be displayed

Detailed DNS packet fields.

FTP Traffic Analysis

Procedure:

1. Connected to FTP service.
2. Logged in using credentials.
3. Captured packets in Wireshark.

Credentials Used:

Username:

msfadmin

Password:

msfadmin

Credential Capture

Using Follow TCP Stream:

The FTP session revealed credentials in plaintext.

Captured Data:

USER msfadmin

PASS msfadmin

230 Login successful

Observation:

FTP transmits credentials without encryption, making it insecure.

Security Recommendation:

Use:

⑩ SFTP

⑩ FTPS

⑩ SCP

instead of FTP.



FTP login credentials captured in plaintext.

7. SYN Flood Simulation using hping3

Objective

Understand how TCP SYN packets can be used in denial-of-service attacks.

Command:

```
sudo hping3 -S 192.168.56.102 -p 80 -c 50
```

Observation:

50 SYN packets were generated and sent to the target machine.

Wireshark captured the SYN packets successfully.

Filter Used:

tcp.flags.syn == 1

```
(kali@kali)-[~]
└─$ sudo hping3 -S 192.168.56.102 -p 80 -c 50
HPING 192.168.56.102 (eth1 192.168.56.102): S set, 40 headers + 0 data bytes

— 192.168.56.102 hping statistic —
50 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(kali@kali)-[~]
```

No.	Time	Source	Destination	Protocol	Length	Info
253	2039.9695019...	192.168.56.1	224.0.0.251	MDNS	82	Standard query 0x0000
254	2039.9695028...	fe80::8680:595c:15b...	ff02::fb	MDNS	102	Standard query 0x0000
255	2039.9735568...	192.168.56.1	224.0.0.251	MDNS	82	Standard query 0x0000
256	2039.9741884...	fe80::8680:595c:15b...	ff02::fb	MDNS	102	Standard query 0x0000
257	2040.7147779...	192.168.56.1	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
258	2041.9186619...	192.168.56.1	224.0.0.251	MDNS	82	Standard query 0x0000
259	2041.9192646...	fe80::8680:595c:15b...	ff02::fb	MDNS	102	Standard query 0x0000
260	2041.9253098...	192.168.56.1	224.0.0.251	MDNS	82	Standard query 0x0000
261	2041.9253105...	fe80::8680:595c:15b...	ff02::fb	MDNS	102	Standard query 0x0000
262	2041.9738481...	192.168.56.1	224.0.0.251	MDNS	82	Standard query 0x0000
263	2041.9738486...	fe80::8680:595c:15b...	ff02::fb	MDNS	102	Standard query 0x0000
264	2041.9800756...	192.168.56.1	224.0.0.251	MDNS	82	Standard query 0x0000
265	2041.9800770...	fe80::8680:595c:15b...	ff02::fb	MDNS	102	Standard query 0x0000
266	2042.3172553...	192.168.56.1	224.0.0.251	MDNS	495	Standard query respons
267	2043.7139515...	192.168.56.1	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
268	2048.8098991...	0.0.0.0	255.255.255.255	DHCP	330	DHCP Discover - Transa
269	2050.3001608...	192.168.56.1	224.0.0.251	MDNS	495	Standard query respons
270	2062.4454108...	0.0.0.0	255.255.255.255	DHCP	330	DHCP Request - Transa
271	2062.4496852...	192.168.56.100	255.255.255.255	DHCP	590	DHCP ACK - Transa
272	2062.4565906...	::	ff02::16	ICMPv6	90	Multicast Listener Rep
273	2062.4709468...	PCSSystemtec_62:f1:...	Broadcast	ARP	42	Who has 192.168.56.101
274	2062.4962181...	PCSSystemtec_62:f1:...	Broadcast	ARP	42	Who has 192.168.56.101
275	2062.5578369...	PCSSystemtec_62:f1:...	Broadcast	ARP	42	Who has 192.168.56.101
276	2062.6218067...	PCSSystemtec_62:f1:...	Broadcast	ARP	42	ARP Announcement for 1
277	2062.9688848...	::	ff02::1:ff49:2df0	ICMPv6	86	Neighbor Solicitation
278	2063.0325168...	::	ff02::16	ICMPv6	90	Multicast Listener Rep

8. Firewall Configuration using iptables

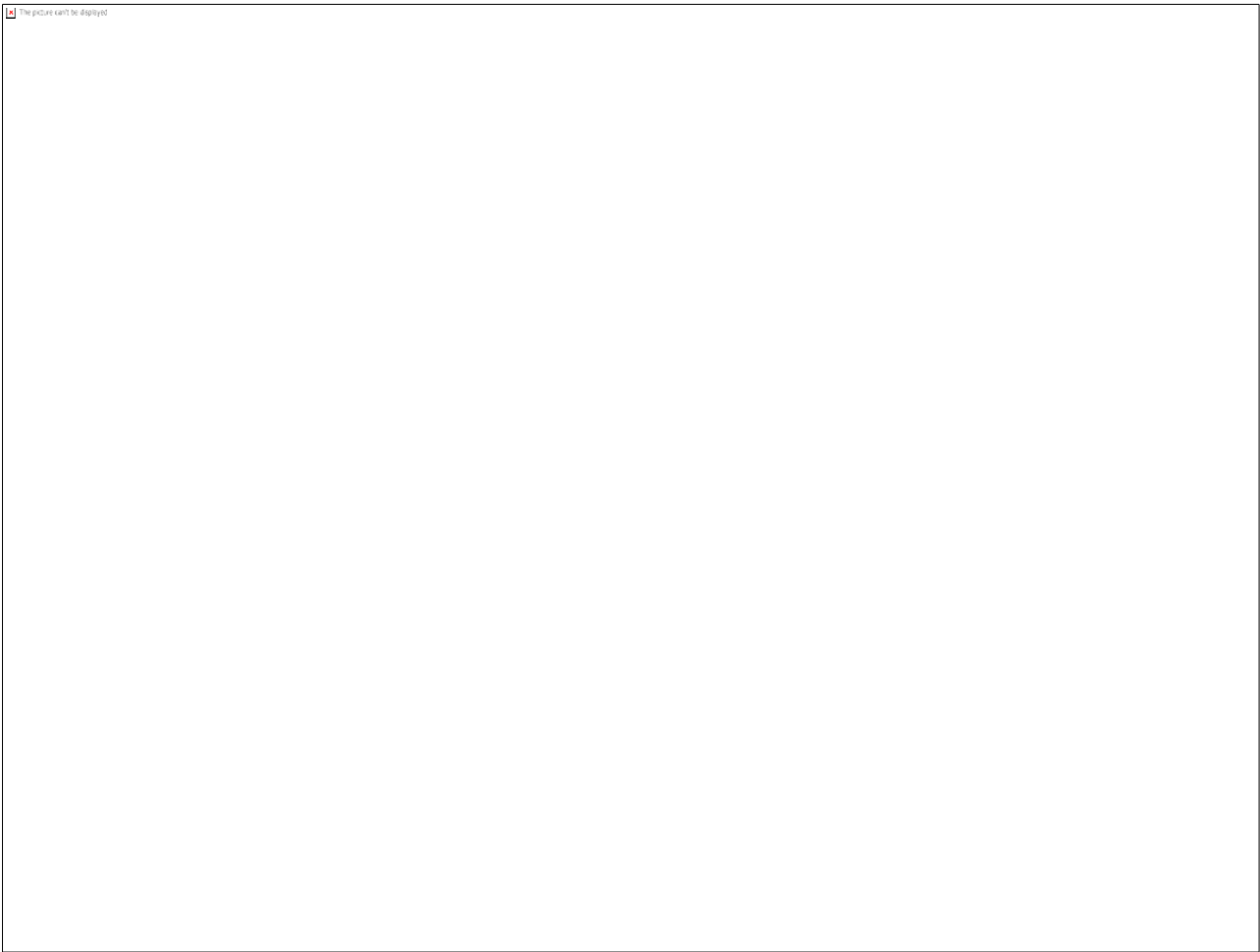
Objective

Demonstrate firewall rule creation and traffic filtering.

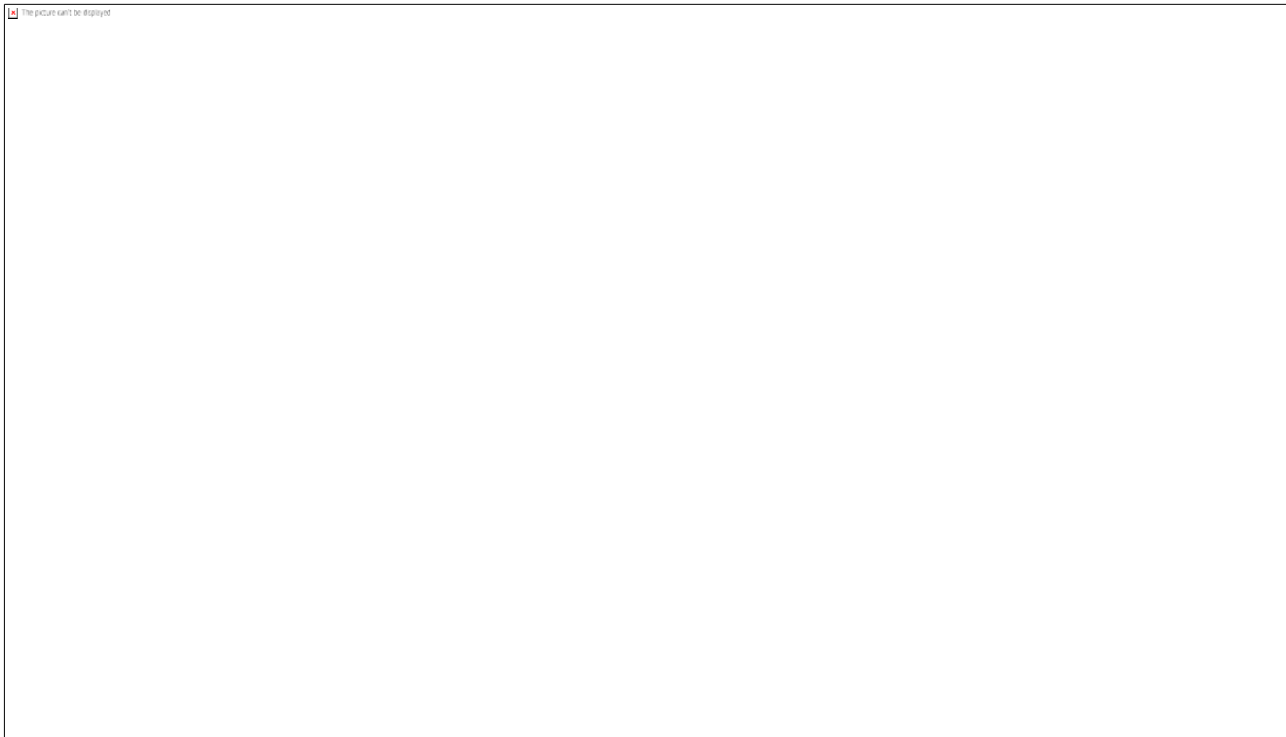
Install hping3

 The picture can't be displayed

Generate SYN Packets



Capture SYN Packets



View Existing Rules

Command:

```
sudo iptables -L
```

Observation:

Default ACCEPT policy was configured.

```
Session Actions Edit View Help
(kali㉿kali)-[~] word.
$ sudo iptables -L
[sudo] password for kali:
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

(kali㉿kali)-[~]
$
```

Add Firewall Rule

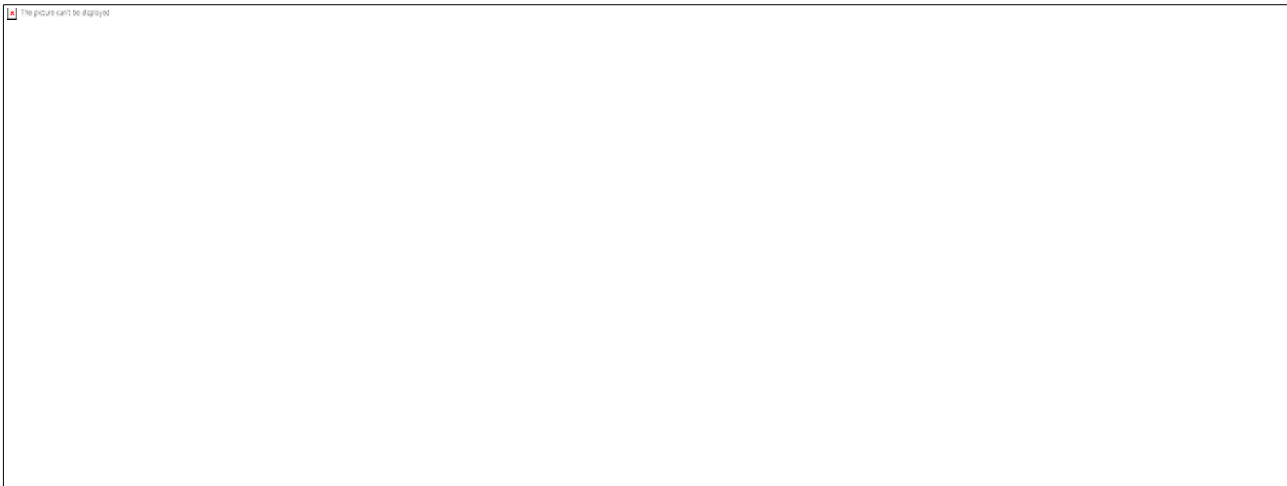
Command:

```
sudo iptables -A INPUT -p tcp --dport 80 -j DROP
```

Observation:

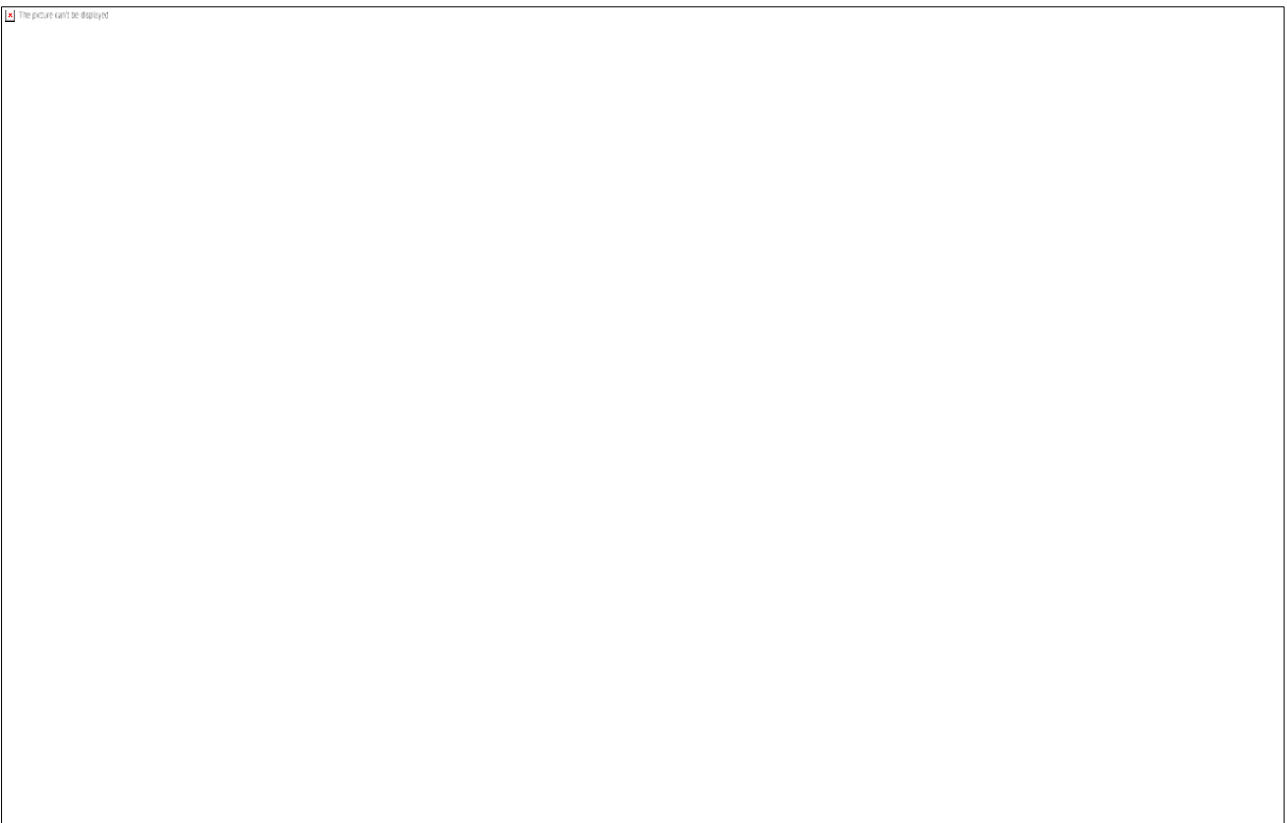
HTTP traffic on port 80 was blocked.





Demonstrate Blocking

When attempting to access the HTTP service, the connection failed due to the firewall rule.



Remove Rule

Command:

```
sudo iptables -D INPUT -p tcp --dport 80 -j DROP
```

Observation:

Normal connectivity was restored.

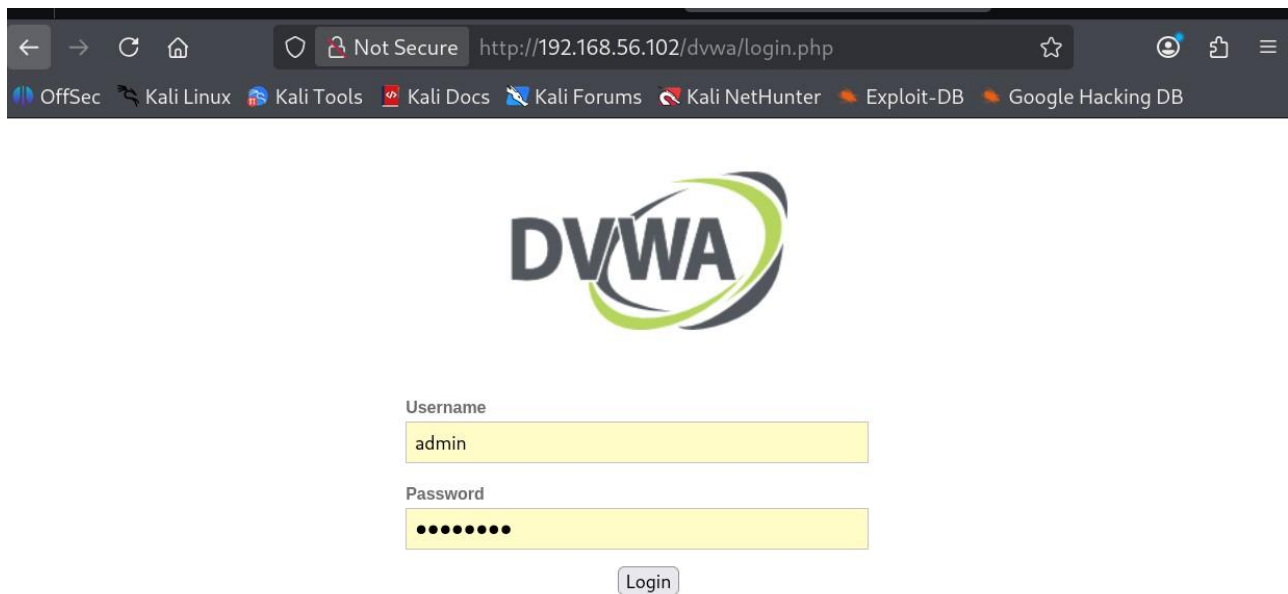
```
msfadmin@metasploitable:~$ sudo iptables -D INPUT -p tcp --dport 80 -j DROP
msfadmin@metasploitable:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
msfadmin@metasploitable:~$ _
```

The picture can't be displayed

Connectivity Restored



← → ↻ 🏠 🔒 Not Secure http://192.168.56.102/dvwa/login.php ☆ ⓘ ⌵ ☰

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

DVWA

Username
admin

Password
●●●●●●

Login

Demonstrating Blocked Port Scan

Create Scan Blocking Rule



Nmap During Block



```
(kali㉿kali)-[~]
$ nmap 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-08 10:28 -0400
Stats: 0:00:03 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
an
SYN Stealth Scan Timing: About 65.95% done; ETC: 10:28 (0:00:01 remaining)
Stats: 0:00:03 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
an
SYN Stealth Scan Timing: About 76.80% done; ETC: 10:28 (0:00:00 remaining)
Stats: 0:00:03 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
an
SYN Stealth Scan Timing: About 82.90% done; ETC: 10:28 (0:00:00 remaining)
Nmap scan report for 192.168.56.102
Host is up (0.043s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
```

Remove Rule



Scan Works Again

The picture can't be displayed.

Google Dorking

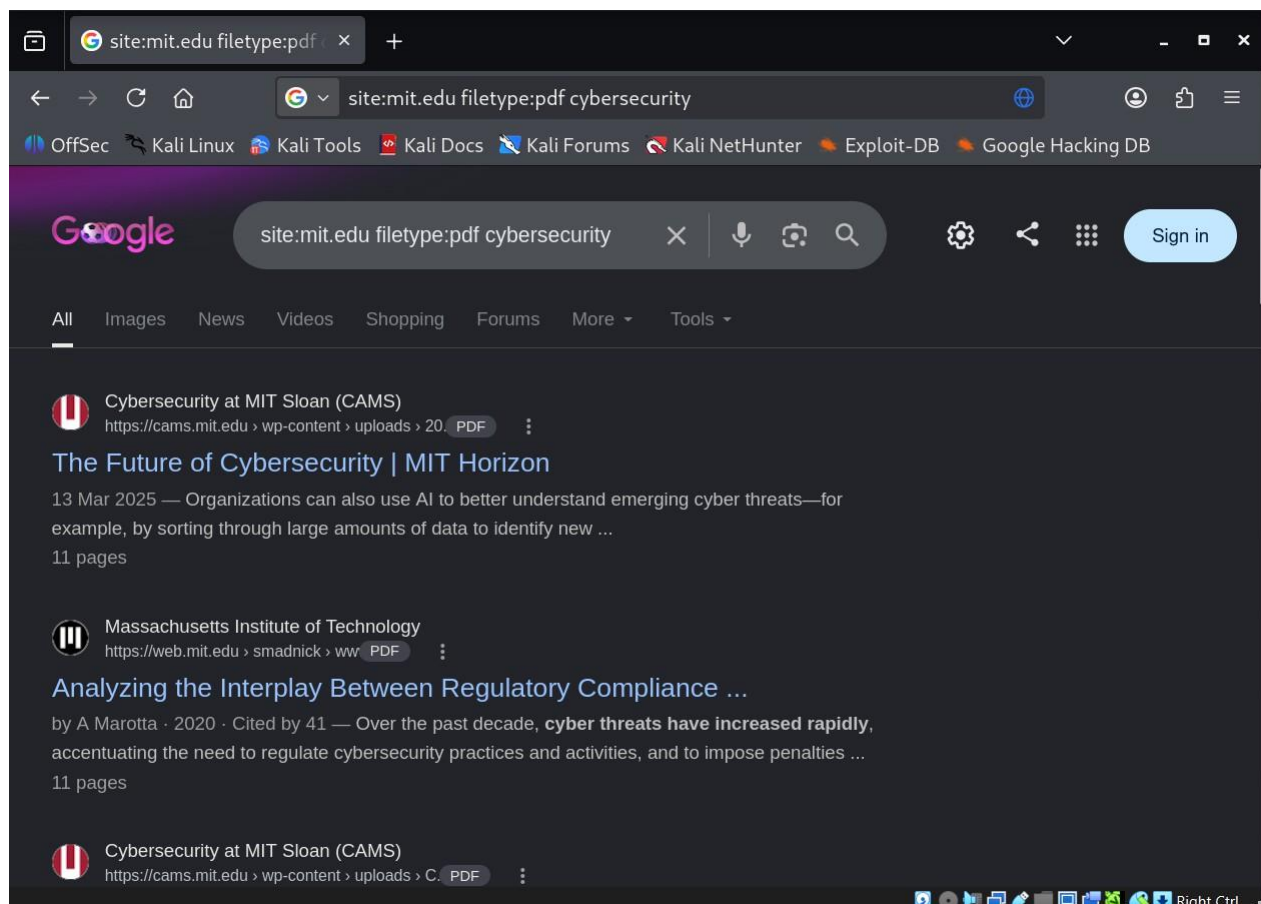
Objective: Discover publicly available information using advanced Google search operators.

Command/Search Query:

site:mit.edu filetype:pdf cybersecurity

Observation:

Google Dorking revealed publicly accessible PDF documents related to cybersecurity. This technique helps security professionals identify exposed information available through search engines.



Shodan Reconnaissance

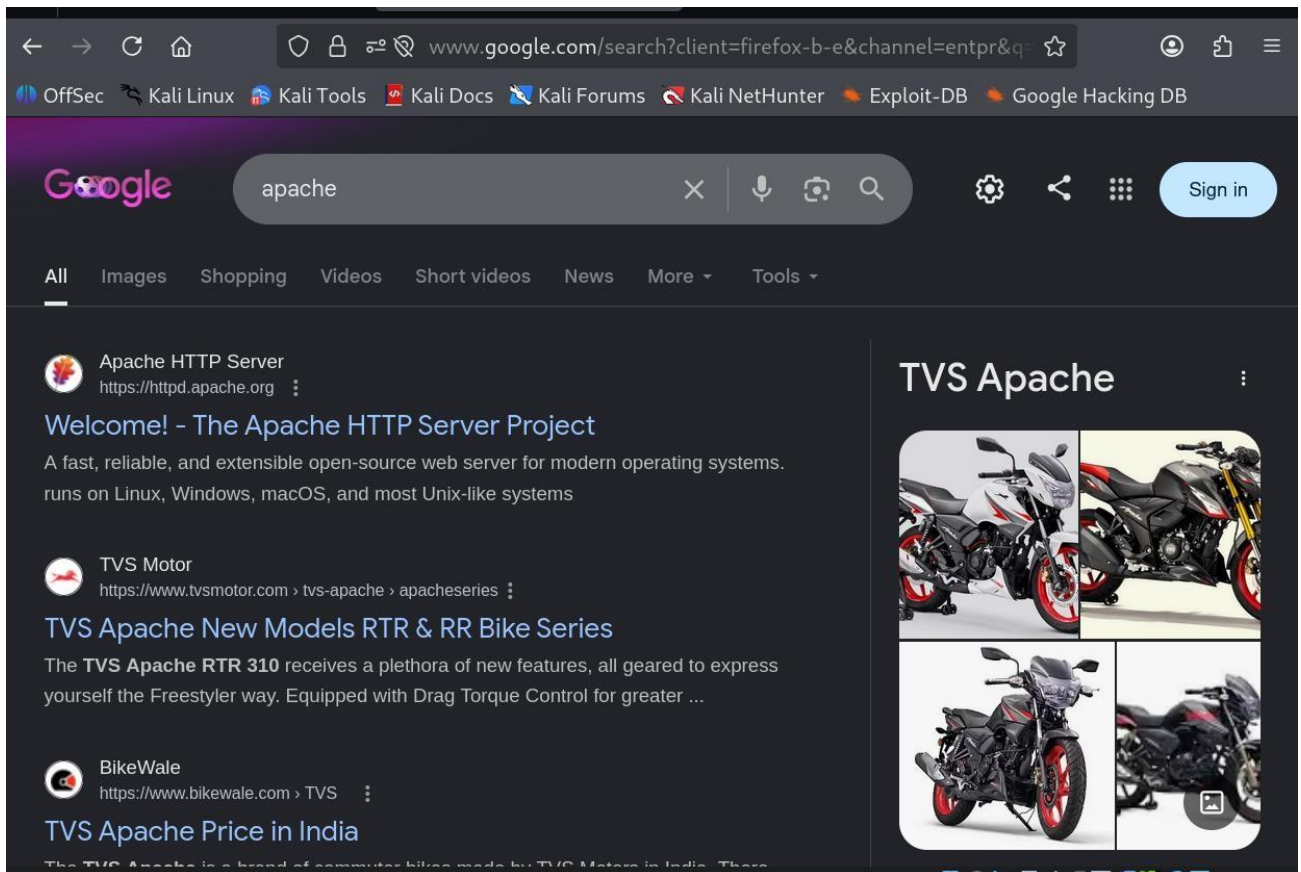
Objective: Identify internet-facing devices and services.

Search Query:

apache

Observation:

Shodan displayed publicly exposed servers and services running Apache web server software. This demonstrates how attackers and defenders can discover internet-facing assets.



Firewall Rule

```
sudo iptables -A INPUT -p tcp --dport 80 -j DROP
```

Observation:

The firewall successfully blocked incoming traffic on port 80.

Port Scan Verification

```
nmap -p 80 192.168.56.102
```

Observation:

The Nmap scan showed that port 80 was filtered after applying the firewall rule, confirming successful traffic blocking.

```
msfadmin@metasploitable:~$  
msfadmin@metasploitable:~$ sudo iptables -A INPUT -p tcp --dport 80 -j DROP  
[sudo] password for msfadmin:  
msfadmin@metasploitable:~$  
msfadmin@metasploitable:~$
```

```
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ nmap -p 80 192.168.56.102  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-14 10:39 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.011s latency).  
  
PORT      STATE SERVICE  
80/tcp    filtered http  
MAC Address: 08:00:27:0C:3B:02 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 3.93 seconds  
  
(kali@kali)-[~]  
$
```

Findings

The assessment revealed several intentionally vulnerable services on the Metasploitable2 machine, including FTP, Telnet, SMB, and outdated web services. Packet analysis demonstrated how insecure protocols expose sensitive information. Nessus identified multiple vulnerabilities, while firewall configuration showed how network access can be controlled effectively.

Conclusion

This task demonstrated practical network reconnaissance, port and service scanning, vulnerability assessment, packet analysis, SYN flood simulation, and firewall configuration. Using tools such as Nmap, Nessus Essentials, Wireshark, hping3, and iptables, security weaknesses were identified and analyzed. The exercise provided hands-on experience with common cybersecurity assessment techniques used in real-world environments.